

Aleksandr Starostin



Personal information

Starostin Aleksandr
Moscow
Nationality: Russian
2006

Education

- MIPT, Radio engineering and computer technology, Applied Mathematics and Physics, 3 course (2024 - now)
- School №41 (ordinary school) (2013 - 2024)

Personality qualities

Fast in studying
Ready to accept criticism
Stubborn in achievement my purposes
Sociable and friendly in communication
Responsible

Interests

Programming
System programming
Cyber security
Soft development
Microcontroller units (MCU)
Informatics, Physics and Math
Electrics
Digital synthesis

Main purpose

Studying and acquisition experience

CV (MARCH 2026)

Here you can check some information about my skills and projects.

SKILLS

It is list of some my skills:

- **Languages:** C, C++ (in process), Go (in process), NASM, AT&T, LaTeX, Python, Verilog and System Verilog.
- **ISA:** x86-64.
- **Tools:** Linux-Ubuntu, Git, Make, Radare2, Perf, Hotspot, Vivado 2019.1.
- **Libraries:** SFML, graphviz.
- **FPGA boards:** Basys 3.
- **MCU boards:** Arduino Nano, ESP32, Raspberry Pi.

BOOKS

Some books were read by me:

- *"The C Programming Language"* by Brian Kernighan and Dennis Ritchie.
- *"Computer Systems: A Programmer's Perspective"* by Randal E. Bryant and David R O'Hallaron.
- *"FPGA programming"* by Frank Bruno.

ADDITIONAL COURSES

1. C programming: way from solution of quadratic equation to compiler (1 term) (by Dedinsky).
2. Assembler programming and learning computer's architecture (2 term) (by Dedinsky).
3. Programming of MCU and robotics (3 term).
4. Learning computer system's architecture (4 term).
5. Learning C++ (5 term) (by Vladimirov).

PROJECTS

All my projects you can check on my [GitHub](#). Here are some of them:

My language and compiler for it

It is compiler for my programming language, which generates assembler file. Backend: assembler for my virtual CPU, NASM (x86-64).

My_printf - analogue of printf from <stdio.h>

In this project NASM is used for writing function 'my_printf' (version of 'printf' without Floating Point numbers). Also 'my_printf' can be linked with program on C.

Hash table

This project is about realization of hash table on C and improving search's performance.

Multiple cache

Multiple cache is hierarchical structure of various caches: LRU, LFU, 2Q, LIRS. The projects is written on C++ and it is in process.

Mandelbrot's model

Mandelbrot's model is math model of fractal geometry with points on complex plane. Intrinsic functions are used for improvement of performance.

Print letter on seven segment

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Code on System Verilog, with can print letter on seven segment. User can choose the letter and position for print by using buttons. This project is initiated on FPGA.

LANGUAGES

Russian



English



GRADES

General MIPT: 7.94/10.00

Informatics: 8.60/10.00

Physics: 8.88/10.00

Math: 7.11/10.00

DEVELOPMENT VECTORS

List of some my purpose on the near future:

- Learning Linux.
- Learning Docker.
- Learning SQL.
- Learning Go and write SSO service.
- Learning distributed computing system.
- Develop in cyber security.
- Learning develop soft products.